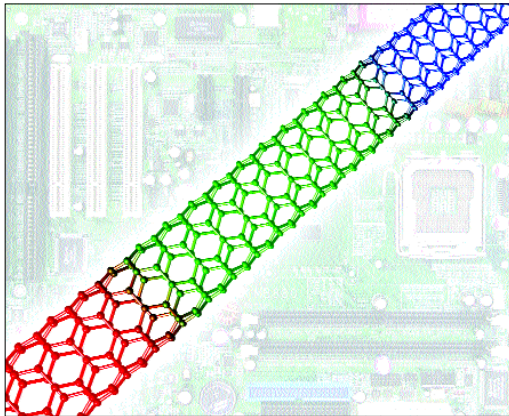




A typical single-walled carbon nanotube. Of course, this is an artist's impression—they aren't really so colourful!

Finally, the resistivity (see *Jargon Buster*) of the interconnects, too, increases as the width of the interconnects gets close to the mean free path of the electrons, which is the average distance the electron travels before encountering a collision. This happens as a result of "electron scattering" in the metal—the electrons getting deflected from their path—and increased resistivity means more power required to pump the current through.

Enter The Carbon Nanotube

Nanotubes are extremely interesting little things. They can be good electrical conductors—meaning they can be used as nanowires (a regular wire at the nanoscale)—and they can also be good semiconductors. They have several other exciting characteristics important to different types of researchers, which we won't be going into.

Constructing a MOSFET using nanotubes is something like replacing the silicon channel with a nanotube. Now, nanotubes can be treated as one-dimensional: they're so thin that, for practical purposes, you can forget that they have a thickness! One result of their one-dimensionality is that there is there are only two directions for the electrons to flow through them—backward or forward, and electrons flowing in the reverse direction ("backscattering") is highly improbable. This means lower resistivity than a three-dimensional wire under similar conditions.

The transport of electrons in a semiconducting nanotube is "ballistic," meaning that the electrons shoot straight forward, without any deflections along their way. This property holds on the scale of a few hundred nanometres—long enough for them to be used in MOSFETs. This means low leakage currents, which means low power dissipation—which, in turn, means low heat. There is also no electromigration. Switching times are faster—which means higher performance. And finally, due to reasons we cannot explain here, there will now be no need to use silicon dioxide as the gate insulator when a nanotube is being used: materials with higher dielectric constants (see *Jargon Buster*) can be used, which again means higher performance.

The picture is more complex than we've presented—so let's take a real-world look at what's been going on in the direction of actually using nanotubes in transistors.

From Lab To Fab

May 2002 was a milestone month for nanotech. IBM announced it had created the highest-performing nanotube transistors ever, and proved that nanotube transistors could outperform state-of-the-art silicon prototypes. The proof centred around transconductance, which is a measure of how well something carries electrical current; the researchers involved reported they had achieved the highest transconductance of any carbon nanotube made thus far, and also that their nanotube transistors had twice the transconductance of the best silicon prototypes. Higher transconductance means transistors can operate faster, and that the resulting IC will be more powerful.

Dr Phaedon Avouris, manager of nanoscale science, IBM Research, is reported to have said, "Carbon nanotubes are already the top candidate to replace silicon when current chip features just can't be made any smaller, a physical barrier expected to occur in about 10 to 15 years."

To go into the details a little, the scientists at IBM made Carbon Nanotube Field-Effect Transistors (CNFETs) in a structure that resembled a regular MOSFET, and that's how they were able to compare the CNFETs with regular silicon transistors. The gate electrodes were above the channel, separated from it by a thin dielectric (see *Jargon Buster*). This was to measure the performance improvements that could be

Jargon Buster

Electron Tunnelling

A quantum mechanical effect, tunnelling is the mechanism by which electrons can cross a barrier they are not "supposed to." Think of it like you facing a thin waterfall: it is a barrier, but you can go through it. More scientifically, tunnelling happens because, at the atomic level, things can be waves and particles at the same time: when an electron tunnels through a barrier, it is acting like a wave.

Electromigration

Electromigration is the result of momentum transfer from electrons moving in an electric field to the ions in the interconnect material. Electromigration can lead to the electrical failure of interconnects.

Resistivity

Resistivity is the measure of how strongly a material opposes the movement of electrical charge through it. If two objects of identical dimensions have different resistivities, the object with the higher resistivity will have a higher resistance.

Dielectric

A dielectric is simply an insulator—usually employed as an insulating medium between conductors.

Dielectric Constant

This constant is the ratio of the charge-holding capacity of a condenser made with a certain dielectric material to the capacity of the same condenser with air as the dielectric. (see *Dielectric*)